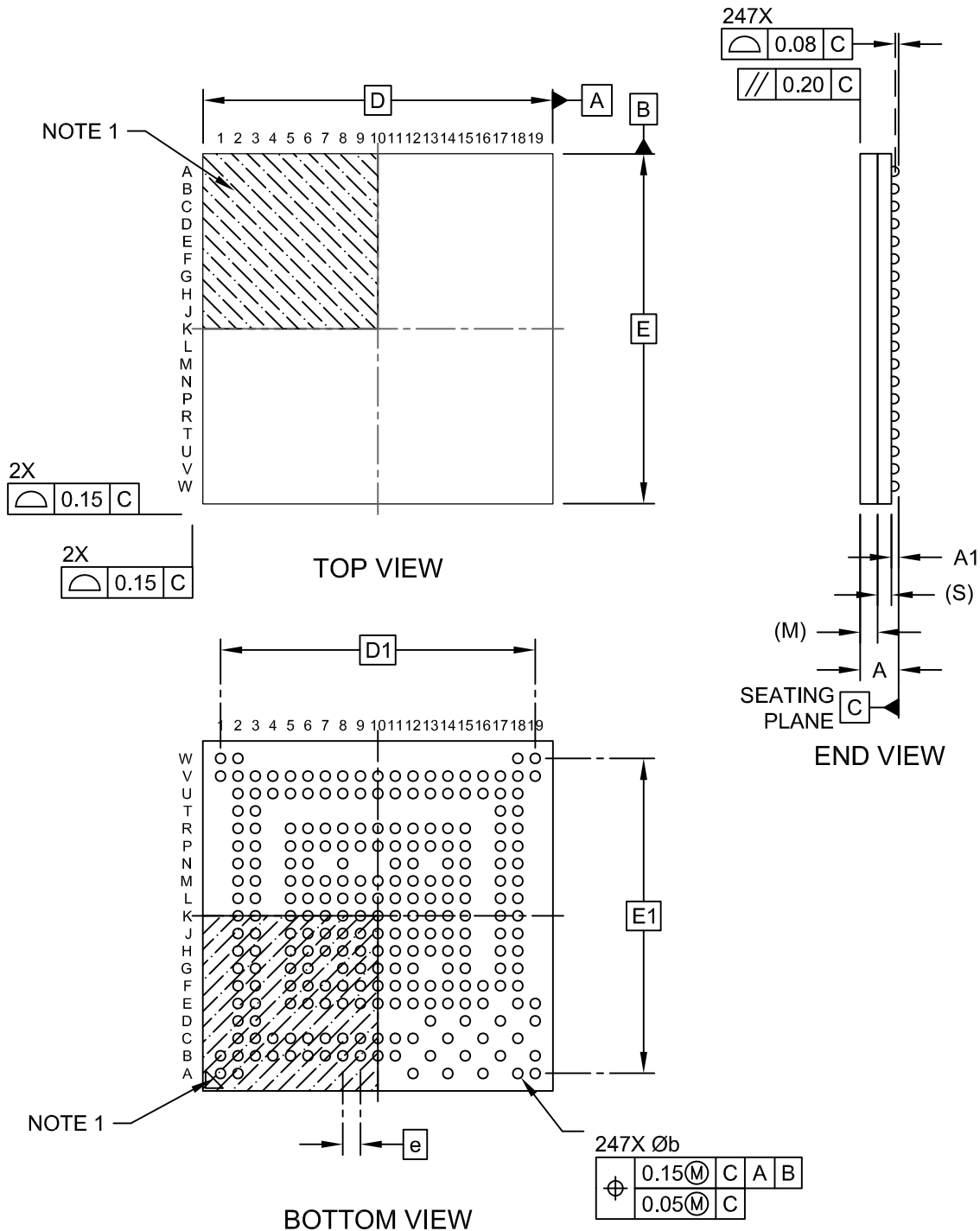


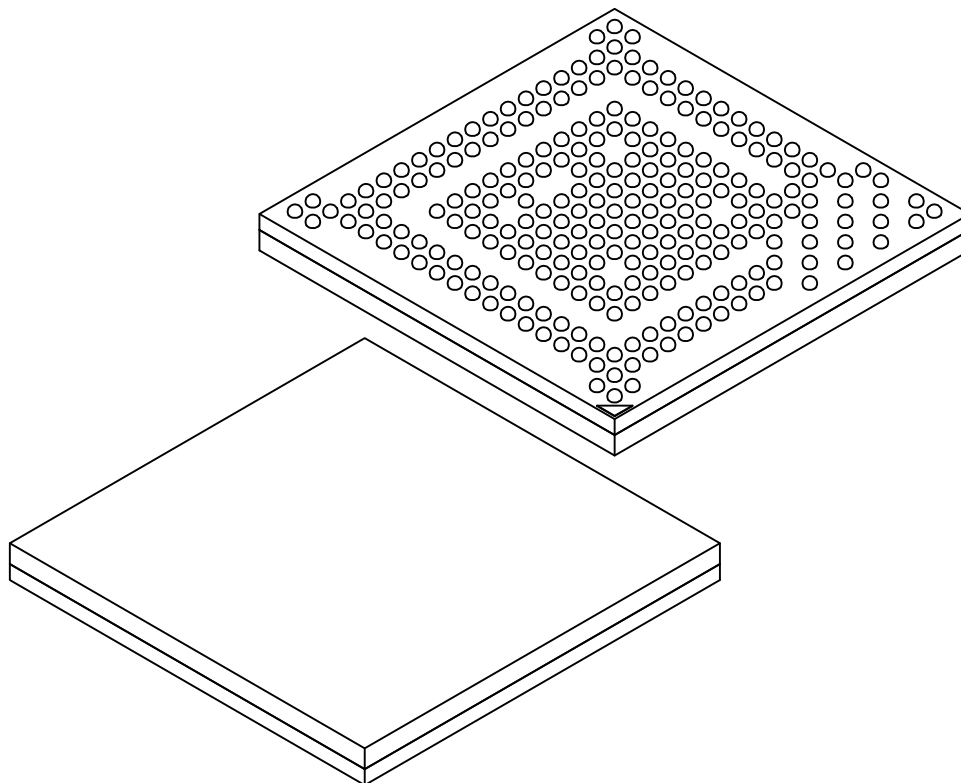
# **247-Lead Thin Fine-Pitch Ball Grid Array Package (B3B) - 10x10x1.1 mm Body [TFBGA]** **Atmel Legacy GPC: CML**

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## 247-Lead Thin Fine-Pitch Ball Grid Array Package (B3B) - 10x10x1.1 mm Body [TFBGA] Atmel Legacy GPC: CML

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



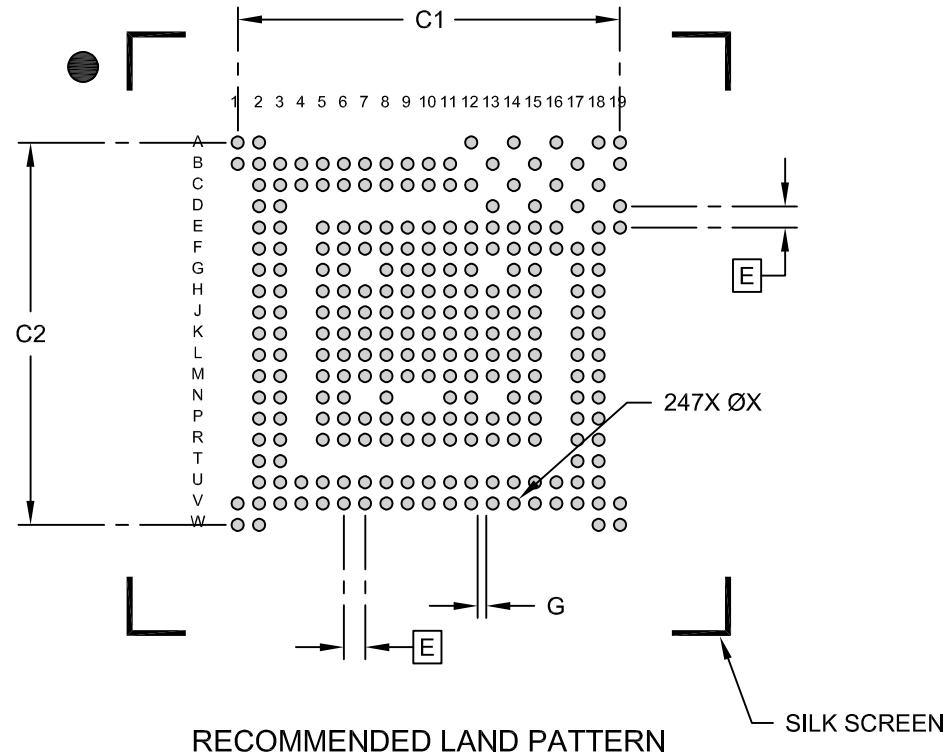
|                     |    | Units | MILLIMETERS |      |      |
|---------------------|----|-------|-------------|------|------|
| Dimension Limits    |    |       | MIN         | NOM  | MAX  |
| Number of Terminals | N  |       | 247         |      |      |
| Pitch               | e  |       | 0.50 BSC    |      |      |
| Overall Height      | A  |       | —           | —    | 1.10 |
| Ball Height         | A1 |       | 0.16        | 0.21 | 0.26 |
| Mold Thickness      | M  |       | 0.50 REF    |      |      |
| Substrate Thickness | S  |       | 0.30 REF    |      |      |
| Overall Length      | D  |       | 10.00 BSC   |      |      |
| Ball Array Length   | D2 |       | 9.0 BSC     |      |      |
| Overall Width       | E  |       | 10.00 BSC   |      |      |
| Ball Array Width    | E2 |       | 9.0 BSC     |      |      |
| Ball Diameter       | b  |       | 0.25        | 0.30 | 0.35 |

### Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
  - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
  - REF: Reference Dimension, usually without tolerance, for information purposes only.

## 247-Lead Thin Fine-Pitch Ball Grid Array Package (B3B) - 10x10x1.1 mm Body [TFBGA] Atmel Legacy GPC: CML

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



| Units                        |    | MILLIMETERS |      |      |
|------------------------------|----|-------------|------|------|
| Dimension Limits             |    | MIN         | NOM  | MAX  |
| Contact Pitch                | E  | 0.50 BSC    |      |      |
| Optional Pad Diameter (X247) | X  |             |      | 0.30 |
| Contact Pad Spacing          | C1 |             | 9.00 |      |
| Contact Pad Spacing          | C2 |             | 9.00 |      |
| Column and Row Spacing       | G  | 0.20        |      |      |

### Notes:

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23136 Rev A